Insurance Claim Fraud Detection Pipeline

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PROBLEM STATEMENT & OBJECTIVE

Problem Statement

Insurance fraud is a significant issue that leads to financial losses for insurance companies. Identifying fraudulent claims manually is inefficient and prone to errors. There is a need for a structured system to process and analyze claim data to detect potential fraud patterns.

Objective

The objective of this project is to build a basic end-to-end data pipeline that:

- Reads insurance claim data from a CSV file
- Cleans and preprocesses the dataset
- Stores the processed data in a PostgreSQL database
- Performs SQL queries for analysis
- Visualizes insights using Power BI

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METHODOLOGY

Data Pipeline Workflow

Implementation Details

- **Data Ingestion:** Dataset dewnloaded and stored locally. Loaded using Pandas.
- **Data Cleaning:** Handled missing values. Created structured dataset (cleaned.csv).
- **Database Integration:** Installed PostgreSQL. Created database insurance_db. Uploaded dataset using Python (ingest.py).
- **SQL Analysis:** Executed queries in pgAdmin: Data preview (SELECT *), Aggregation (GROUP BY), Basic ranking using window function.
- **Visualization:** Connected Power BI to PostgreSQL. Loaded table public.claims. Created charts: Fraud vs Non-Fraud, Fraud by Region, Vehicle-based analysis.

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TECH STACK & TASK MAPPING

Tech Stack

- Python (Pandas)
- PostgreSQL (pgAdmin)
- SQL
- Power BI
- Jupyter Notebook

Task Mapping (As per Project Helper)

Task 3: Python Basics
Read CSV file using Pandas
Cleaned missing values

Task 4: Advanced Python
Created Python script (ingest.py) for data upload

Task 7: Advanced SQL
Used GROUP BY and window functions

Task 11: Data Ingestion
Built pipeline from CSV to PostgreSQL

Task 30: Final Project
Implemented end-to-end pipeline:
ingestion → storage → query → visualization

5 RESULTS & DASHBOARD

Key Insights

- Distribution of fraud vs non-fraud claims
- Region-wise claim patterns
- Vehicle characteristics influencing claims

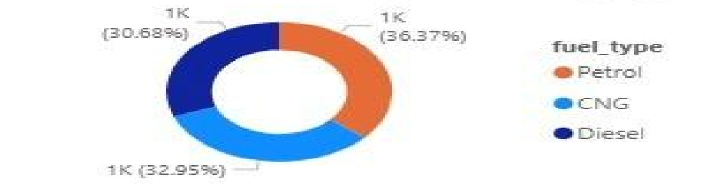
Power BI Dashboard

INSURANCE CLAIM FRAUD DETECTION DASHBOARD

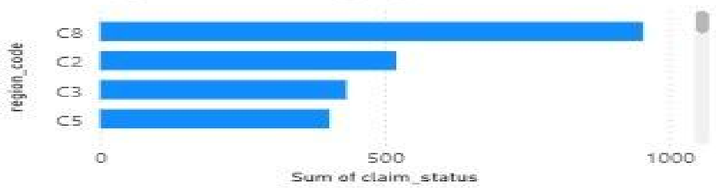
Total Claims	Fraud Claims	Legitimate Claims
58.592K	3.748K	54.844K

Fraud vs. Legitimate Ratio

Fraud Distribution by Fuel Category



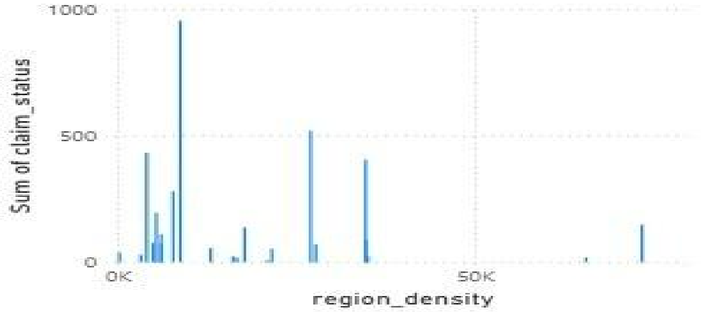
High-Risk Geographic Clusters



Fraud Probability by Vehicle Seniority

model	Engine displacement	Fraud Incident Count
M1	11898608	918
M10	1445964	73
M11	435237	15
M2	1292760	80
M3	2370627	128
M4	20928874	901
M5	2392206	116
M6	16489872	939
M7	3519180	201
M8	4164654	244
M9	3166772	133
Total	68104754	3748

Claim Density vs. Urbanization Level



CONCLUSION & FUTURE WORK

Conclusion

This project demonstrates a basic data engineering workflow where raw data is processed, stored, analyzed, and visualized. It helps in understanding how structured pipelines can support fraud detection analysis.

Future Improvements

- Implement machine learning for fraud prediction
- Use PySpark for large-scale processing
- Add real-time data streaming
- Deploy pipeline on cloud platforms